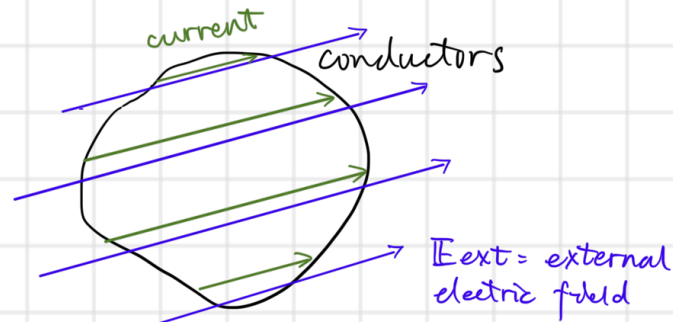
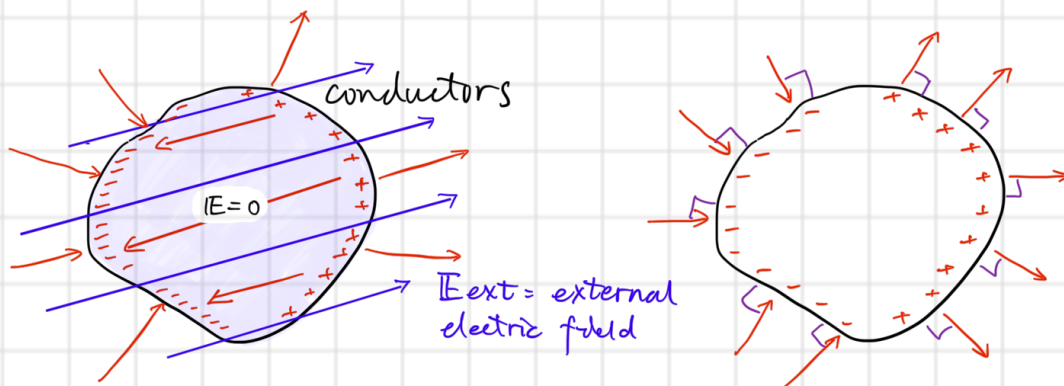


Conductors in electric equilibrium.

A good conductor contains a huge number of free electrons that are not bound to atoms and move around freely.



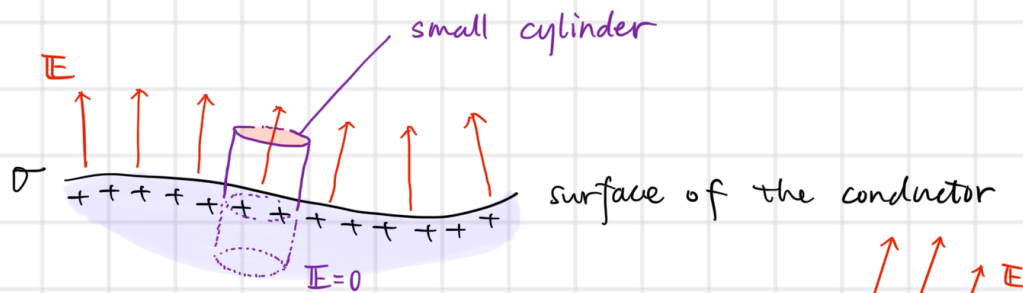
Put the conductor in E_{ext} , current flows.
Wait till the current stops
 $\Rightarrow$ Reach electric equilibrium



$$E = E_{\text{ext}} + E_{\text{created by charges inside conductor}}$$

- $E = 0$ everywhere inside the conductor in electric equilibrium. (Otherwise current is present where $E \neq 0$)
- All charges are located on the surface of the conductors (because isolated charges inside will create E around them)
- E is perpendicular to the surface of the conductor, right outside the surface (Otherwise there will be current on the surface.)

Apply Gauss' Law in this situation:



$$\Phi_E = E_{\text{right outside of the conductor}} \cdot A$$

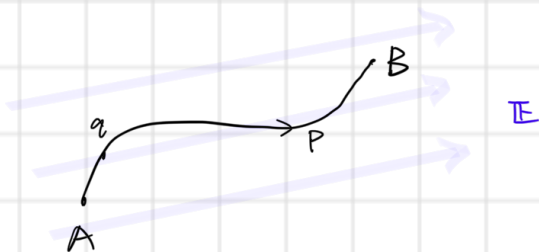
$$(\text{Gauss' Law}) = \frac{\sigma \cdot A}{\epsilon_0} \Rightarrow E_{\text{right outside of the conductor}} = \frac{\sigma}{\epsilon_0}$$

(Similar to the infinite plane of charge, but without $\frac{1}{2}$)

Electric potential

- Coulomb force is conservative (for the same reason as Newtonian Gravity is)

Consider moving a point charge q in electric field.



$$\begin{aligned} W_P &= \int_P q \cdot \vec{E} \, ds \\ &= -\Delta U \\ &= -U(B) + U(A) \end{aligned}$$

where U is a potential function.

Define the **electric potential difference** by

$$\Delta V = \frac{\Delta U}{q} = - \int_A^B \vec{E} \cdot d\vec{s}$$

Unit for ΔV : $J/C = V$ (volt)

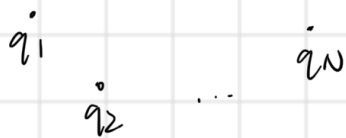
To define V itself, we choose a point p_0 and set

$$V(p) = - \int_{p_0}^P \vec{E} \cdot d\vec{s}$$

In many cases, we choose $p_0 = \infty$.

$$\vec{E} = -\nabla V$$

Ex. Poisson charges



$$E(r) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{q_i}{|r-r_i|^3} (r-r_i)$$

$$V(r) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{q_i}{|r-r_i|}$$

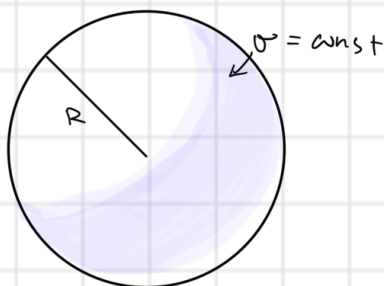
$$E = -\nabla V.$$

$$V(r=\infty)=0 \Rightarrow \text{Reference point } p_0 = \infty.$$

For continuous charge distribution,

$$V(r) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(r') dV'}{|r-r'|}$$

Ex. Spherical shell of charge



We know E already outside the sphere, E is the same as that of point charge

$$Q = 4\pi R^2 \sigma$$

$$V(r) = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{Q}{r} = \frac{R^2 \sigma}{\epsilon_0 r}, & r > R \\ \frac{R\sigma}{\epsilon_0}, & r \leq R \end{cases}$$

≠ Stokes Theorem

$$\oint_{\partial S} \mathbf{V} \cdot d\mathbf{s} = \int_S (\nabla \times \mathbf{V}) \cdot d\mathbf{A}$$

$\mathbf{E}$ is conservative $\Leftrightarrow \nabla \times \mathbf{E} = 0$

$$\nabla \times \mathbf{E} = 0$$

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad \text{Poisson eq.}$$

$$\mathbf{E} = -\nabla V.$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$